

Protocol for methyl-C sequencing of single cell nuclei: snmC-seq

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Reference: "Robust single-cell DNA methylome profiling with snmC-seq2"
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Center for Epigenomics of the Mouse Brain Atlas (CEMBA)

Part of the NIH BRAIN Initiative Cell Census Network (BICCN)

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snmC-seq2 protocol

Protocol to bisulfite convert and generate 3,072 single-cell methylome libraries in one experiment (8x 384-well plates) from mouse or human frozen brain tissue.

Materials

General reagents

- FACS sorted single-nuclei into 384 well PCR plate
- HyClone™ HyPure™ Molecular Biology Grade (MB) Water (GE Life Sci. cat. no. SH30538.03)
- 200-Proof (100%) Ethanol (Koptec cat. no. V1001)

Collection of single nuclei by Fluorescence-activated Cell Sorting (FACS)

- M-Digestion Buffer (Zymo cat. no. D5021-9)
- Proteinase K (Zymo cat. no. D3001-2-D)
- Proteinase K Storage Buffer (Zymo cat. no. D3001-2-B)
- Unmethylated Lambda DNA (Promega cat. no. D1521, 100pg/μL)

Bisulfite conversion

- CT Conversion Reagent (Zymo cat. no. D5003-1)
- M-Solubilization Buffer (Zymo cat. no. D5021-7)
- M-Dilution Buffer-Gold (Zymo cat. no. D5006-2)
- M-Reaction Buffer (Zymo cat. no. D5021-8)
- M-Binding Buffer (Zymo cat. no. D5021-7)
- M-Wash Buffer (Zymo cat. no. D5040-4)
- M-Desulphonation Buffer (Zymo cat. no. D5040-5)
- M-Elution Buffer (Zymo cat. no. D5007-6)

Random primers

- HPLC purified Random primers were ordered from Integrated DNA Technologies (IDT)
P5L_AD002_H
/5SpC3/TTCCCTACACGACGCTCTTCCGATCTCGATGT(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)
P5L_AD006_H
/5SpC3/TTCCCTACACGACGCTCTTCCGATCTGCCAAT(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)
P5L_AD008_H
/5SpC3/TTCCCTACACGACGCTCTTCCGATCTACTTGA(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)

P5L_AD010_H

/5SpC3/TTCCCTACACGACGCTCTTCCGATCTTAGCTT(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)

P5L_AD001_H

/5SpC3/TTCCCTACACGACGCTCTTCCGATCTATCACG(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)

P5L_AD004_H

/5SpC3/TTCCCTACACGACGCTCTTCCGATCTTGACCA(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)

P5L_AD007_H

/5SpC3/TTCCCTACACGACGCTCTTCCGATCTCAGATC(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)

P5L_AD012_H

/5SpC3/TTCCCTACACGACGCTCTTCCGATCTCTTGTA(H1:33340033)(H1)(H1)(H1)(H1)(H1)(H1)(H1)

Sera-Mag Solid Phase Reversible Immobilization (SPRI) beads

- Sera-Mag SpeedBeads Magnetic Carboxylate Modified (GE Healthcare cat. no. 45152105050250)
- Poly(ethylene glycol) PEG 8000 (Sigma cat no. 89510-250G-F)
- TE buffer pH=8.0 (Ambion cat. no. AM9858)
- 5M NaCl
- 1M Tris-HCl pH=8.0
- 0.5M EDTA pH=8.0
- AMPure XP beads (Beckman Coulter cat no. A63881)
- 100 bp DNA ladder (New England Biolabs cat no. N3231L)

snmC-seq2 library preparation

- Blue Buffer (10x) (Enzymatics cat. no. P7010-HC-L)
- Klenow Exo- (50U/μL) (Enzymatics cat. no. P7010-HC-L)
- Deoxynucleotide Solution Mix (10mM each dNTP) (NEB cat. no. N0447L)
- Exonuclease I (20U/μL) (Enzymatics cat. no. X8010L)
- Shrimp Alkaline Phosphatase (rSAP) (1U/μL) (NEB cat. no. M0371L)
- Sera-Mag SPRI beads
- M-Elution Buffer (Zymo cat. no. 5007-6)
- EB buffer (Qiagen cat. no. 19086)
- Accel-NGS® Adaptase™ Module (Swift Bio cat. no. 330384)
- P5 Indexing Primer (IDT custom DNA oligo, standard desalted)
- P7 Indexing Primer (IDT custom DNA oligo, standard desalted)

- KAPA HiFi HS RM (Kapa cat. no. KK2602)
- Qubit dsDNA BR Assay Kit (Thermo Fisher cat. no. Q32850)

Equipment

- 384-Well Hardshell PCR Plate Clear, 20 Pcs/pk (Thermo Fisher cat. no. 4483285)
- 96-Well Hardshell PCR Plate GPLE, 20 Pcs/pk (Thermo Fisher cat. no. 4483348)
- Zymo-Spin 384 Well Plate, 2 pack (Zymo cat. no. C2012)
- 2.0mL 96-well Deep Well Polypropylene Plate, Sterilized (USA-SCI. cat. no. 1896-2110)
- Reservoir Single Well 96 Bottom High Profile, Clear (Axygen cat. no. RES-SW96-HP)
- Reservoir Single Well 96 Bottom Low Profile, Clear (Axygen cat. no. RES-SW96-LP)
- 15mL Centrifuge Tubes (Olympus cat. no. 28-103)
- 50mL Centrifuge Tubes (Olympus cat. no. 28-106)
- 1.5µL Eppendorf Tubes (Thermo Fisher cat. no. 02-681-320)
- 300µL 8-Strip Tubes
- Microamp Clear Adhesive Film, 100pc (Thermo Fisher cat. no. 4306311)
- Microporous Film, -20C to 80C, 50p, sterile (USA-SCI. cat. no. 2920-1010)
- Speedball Deluxe Soft Rubber Brayer, 4 inches (Statesville N.C.)
- 37°C Incubator
- 384-well and 96-well Compatible Thermocycler
- DynaMag™-96 Side Magnet (Thermo Fisher cat. no. 12331D)
- DynaMag™-2 Magnet (Thermo Fisher cat. no. 12321D)
- Allegra 25R Centrifuge with S5700 2x96 Swinging Bucket Rotor (Beckman Coulter cat. no. 368954)
 - All centrifuge steps at 5000xg for 5 minutes at room temperature unless otherwise specified
- (Optional) Axxygen DITI 50µL STE.FIL. Robotics Tips (Thermo Fisher cat. no. EVF-50-RS)
- (Optional) Axxygen DITI 180µL STE.FIL. Robotics Tips (Thermo Fisher cat. no. EVF-180-RS)
- (Optional) TECAN Freedom Evo 100 Base Unit with MultiChannel™ Arm MCA 96
- (Optional/Required for TECAN use) Collection Spacer/Adapter, Type 2, 26.5mm, Te-VacS (Tecan US Inc. cat. no. 10760662)

Reagent Setup

Ethanol, 80%

To 40mL of 200-proof Ethanol, add 10 μ L MB water in a 50mL tube. Keep solution sealed when not in use. Prepare ~6 tubes fresh before each library preparation.

Digestion Buffer Plates (for FACS sorting)

In a 50mL tube, combine 15mL M-Digestion Buffer (incubate for 10 minutes at 37°C to dissolve precipitate) with 14mL MB water. Resuspend one tube Proteinase K with 1mL Proteinase K Resuspension Buffer and add 1mL to the digestion buffer mix. Add 10 μ L λ DNA to aid determination of bisulfite conversion efficiency. Vortex to mix. Aliquot 2 μ L per well to 20 384-well plates to use for FACS sorting of stained nuclei (TECAN script). Plates can be stored at 4C.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - 2uL_Digestion_Buffer.

Sera-Mag Solid Phase Reversible Immobilization (SPRI) Beads

1. Mix Sera-Mag SpeedBeads and transfer 1mL to a 1.5ml tube.
2. Place SpeedBeads on a magnetic stand until clears and carefully remove the supernatant. Wash the beads twice with 1ml of TE. For each wash, remove the tube from the magnet and mix by inversions. Resuspend the washed beads in 1ml of TE.
3. Add 9g of PEG 8000 to a new 50ml sterile conical tube.
4. Add 10ml of 5M NaCl to the 50ml tube.
5. Add 500 μ l of 1M Tris-HCl pH=8.0 and 100ul of 0.5M EDTA pH=8.0 to the 50ml tube.
6. Mix until all dissolves into solution.
7. Add 1ml of resuspended SpeedBeads to the 50ml tube and fill the volume with MB water.
8. Test against AMPure XP beads using 100bp DNA ladder.

CT Conversion Reagent

Add 7.9 mL M-Solubilization Buffer and 3 mL M-Dilution Buffer to a bottle of CT Conversion Reagent. Shake vigorously at room temperature to fully dissolve before adding 1.6 mL M-Reaction Buffer.

M-Wash Buffer

Add 288mL 200-proof Ethanol to four bottles of M-Wash buffer. Invert to mix. Make fresh each time. Extra buffer can be stored at room temperature.

Random Primer solution

35µL random primer stock (100µM) in 7mL M-Elution buffer (500nM final primer concentration). Aliquot to eight 96-well plates.

Random Priming Master Mix

Combine 3600µL Blue Buffer (10x), 900µL Enzymatics Klenow exo- (50U/µL), 1800µL dNTP (10mM each), and 11700µL MB water in a 15mL tube and vortex to mix. Aliquot 180µL of master mix into each well of one 96-well plate. Keep sealed on ice until use. Do not store.

Exo/rSAP Master Mix

Combine 3600µL Exonuclease 1 (20U/µL) with 1800µL rSAP (1U/µL) in a 15mL tube and vortex to mix. Aliquot 53µL master mix to each well on one 96-well plate. Keep sealed on ice until use. Do not store.

Adaptase Master Mix (using Accel-NGS® Adaptase™ Module)

In a 15mL tube, to 1912.5µL M-Elution buffer, add 900µL Buffer G1, 900µL Reagent G2, 562.5µL Reagent G3, 225µL Enzyme G4, and 225µL Enzyme G5. Aliquot 295µL to 2 sets of 8-strip tubes (16 tubes total). Keep sealed on ice until use. Do not store.

PCR Primer Mix

Sequences of indexing primers with unique dual barcodes are provided in Supplementary Table 2. Each PCR Primer Mix contains a P5L indexing primer (600 nM) and a P7L indexing primer (1 µM).

Procedure

Bisulfite Conversion

Timing ~ 5h

1. Add 15µL CT conversion reagent to each well of 384-well plate. Pipette up and down for 8 times to mix the sample.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *15uL_Conversion_Buffer*

2. Seal the plate with adhesive film and quick spin for 10s at 2,000xg at room temperature. Place the plate in a thermocycler and run the following program:

98°C	8 min
64°C	3.5hrs

4°C Hold

3. Load 80µL M-Binding buffer to each well of 384-Well DNA Binding Plate.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *80uL_Binding Buffer*

4. Transfer bisulfite conversion reactions to Zymo-Spin 384 Well DNA Binding Plate. Pipette up and down for 8 times to mix the sample. Place the 384-Well DNA Binding Plate on a 2.0mL 96-well Deep Well Plate and centrifuge for 5 min at 5,000g. Discard the flow through by decanting.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *17uL_Conversion_Rxn_To_Binding_Buffer*.

5. Add 100µL M-Wash Buffer to each well of 384-Well DNA Binding Plate. Centrifuge for 5 min at 5,000g and discard the flow through by decanting.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *100uL_M_Wash_Buffer*.

6. Add 50µL M-Desulphonation Buffer to each well of 384-Well DNA Binding Plate. Incubate at room temperature for 15 min. Centrifuge for 5 min at 5,000g and discard the flow through by decanting.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *50uL_Desulphonation_Buffer*.

7. Add 100µL M-Wash Buffer to each well of 384-Well DNA Binding Plate. Centrifuge for 5 min at 5,000g and discard the flow through by decanting.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *100uL_M_Wash_Buffer*.

8. Repeat step 7.

9. Place 384-Well DNA Binding Plate on new 384-well PCR plate. Add 7µL Random Primer Solution. Each quadrant of 384-well plate is barcoded with a distinct indexed random primer (Fig. 1). Every two 384-well plates receive a complete set of all eight

indexed random primers. Incubate for 5 min at room temperature. Centrifuge for 5 min at 5,000g and discard the 384-Well DNA Binding Plate.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *7uL_Primer_Elution*.

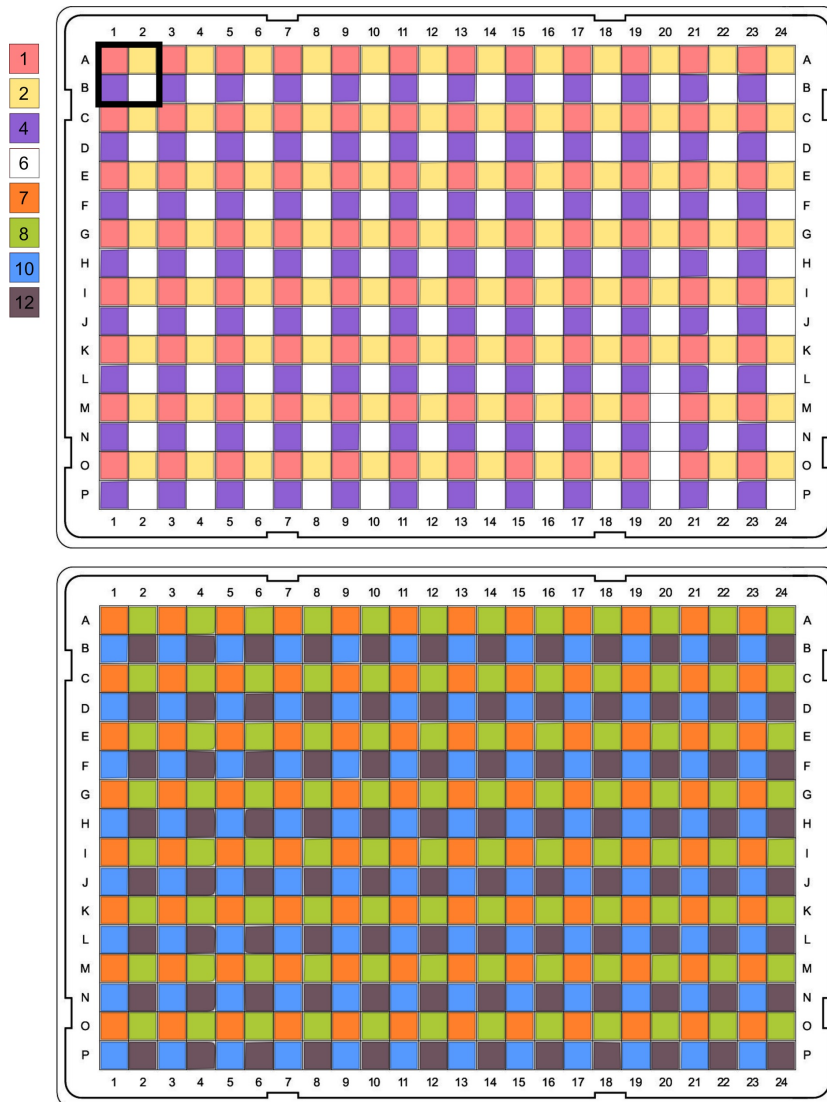


Figure 1. Schematics of single-cell barcoding with indexed random primers. Using a TECAN Freedom Evo 100 with MultiChannel Arm MCA 96, each indexed random primer is added into a quadrant of 384-well DNA Binding Plate. The color of each quadrant indicates the random primer barcode.

10. Seal the plate with adhesive film and store at -20°C for up to 1 week.

Random-primed DNA synthesis

Timing ~2h

1. Denature the samples by placing 384-well PCR plate on a thermocycler and run the following program.

95°C 3 min

Immediately place the plate on ice for 2 minutes.

2. Add 5µL Random Priming Master Mix to each well of the 384-well PCR plate. Vortex and quick spin for 10s at 2,000xg.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *5uL_Random_Priming_Mix*.

3. Place the plate in a thermocycler and run the following program:

4°C 5 min
25°C 5 min
37°C 60 min
4°C Hold

Inactivation of free primers & dNTP

4. Add 1.5µL Exo/rSAP Master Mix to each well of the 384-well PCR plate. Vortex to mix the samples and quick spin for 10s at 2,000xg.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *1uL_Exo_rSAP_MM*.

5. Place the plate in a thermocycler and run the following program

37°C 30 min
4°C Hold

Sample clean-up

6. Add 73.6µL (0.8x) SPRI Beads to each well of four clean 96-well PCR plates.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_Add_To_Clean_96well*.

7. Transfer samples from two 384-well plates to each 96-well plate. The eight quadrants (from two 384-well plates) barcoded with distinct indexed random primers are combined. Mix the samples by vortexing and incubate for 5 minutes at room temperature, then quick spin for 10s at 2,000xg.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_Transfer_Lib_384well_To_96well*.

8. Place 96-well plates on DynaMag™-96 Side Magnet, let stand until solution in wells is clear of beads (~5 minutes). Remove supernatant and wash beads 3x with 150µL fresh 80% EtOH. Remove all EtOH, remove plates from magnet, and let beads dry at room temperature. DO NOT overdry beads.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_2Plate_150uL_EtOH_Wash*.

9. Add 10µL EB buffer and resuspend beads by pipet. Vortex and incubate for 5 minutes at room temperature, then quick spin for 10s at 2,000xg. Place back on magnet and let stand until solution is clear of beads (~5 minutes).

10. Remove 10uL supernatant to a clean 96-well PCR plate.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *10ul_Pre_Adaptase_Transfer*.

Adaptase reaction

11. Denature the samples by placing 96-well plates on a thermocycler and run the following program.

95°C 3 min

Immediately place the plate on ice for 2 minutes.

12. Add 10.5µL Adaptase Master Mix to each well of the 96-well PCR plate. Vortex and quick spin for 10s at 2,000xg.

13. Place the plate in a thermocycler and run the following program:

37°C 30 min
95°C 2 min

4°C Hold

Library amplification

14. Add 5 μ L PCR Primer Mix.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *5uL_PCR_Primer*.

15. Add 25 μ L 2x KAPA HiFi Mix. Vortex and quick spin for 10s at 2,000xg.

16. Place the plate in a thermocycler and run the following program:

- | | | |
|-----------------------|------|-------|
| a. | 95°C | 2 min |
| b. | 98°C | 30s |
| c. | 98°C | 15s |
| d. | 64°C | 30s |
| e. | 72°C | 2 min |
| Go to step c 14 times | | |
| f. | 72°C | 5 min |
| g. | 4°C | Hold |

Library clean-up

17. *Sample cleanup.* Add 40 μ L (0.8x) SPRI Beads to each well of four 96-well PCR plates. Transfer contents of one 96-well plate to another to combine four 96-well plates to two 96-well plates. Vortex and incubate for 5 minutes at room temperature, then quick spin for 10s at 2,000xg.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_Add_Beads_And_Combine_96well*.

18. Place 96-well plates on DynaMag™-96 Side Magnet, let stand until solution in wells is clear of beads (~5 minutes). Remove supernatant and wash beads 2x with 150 μ L fresh 80% EtOH. Remove all EtOH, remove plate from magnet and let beads dry at room temperature. DO NOT overdry beads.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_2Plate_150uL_EtOH_Wash* (stop script after two wash steps).

19. Add 25 μ L EB and resuspend beads by pipet. Vortex and incubate for 5 minutes at room temperature, then quick spin for 10s at 2,000xg. Place back on magnet

and let stand until solution is clear of beads (~5 minutes). Remove 25uL supernatant to a clean 96-well PCR plate.

20. Add 20μL (0.8x) SPRI Beads to each well of two 96-well PCR plates. Transfer contents of one 96-well plate to another to combine two 96-well plates to one 96-well plate. Vortex and incubate for 5 minutes at room temperature, then quick spin for 10s at 2,000xg.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_Add_Beads_And_Combine_96well*.

21. Place 96-well plate on DynaMag™-96 Side Magnet, let stand until solution in wells is clear of beads (~5 minutes). Remove supernatant and wash beads 2x with 150μL fresh 80% EtOH. Remove all EtOH, remove plate from magnet, and let beads dry at room temperature. DO NOT overdry beads.

(Optional) This step can be automated using TECAN Freedom Evo 100 with script - *BP_1Plate_150uL_EtOH_Wash*.

22. Add 20μL EB buffer and resuspend beads by pipet. Vortex and incubate for 5 minutes at room temperature, then quick spin for 10s at 2,000xg. Place back on magnet and let stand until solution is clear of beads (~5 minutes). Combine 20μL eluent from all wells in each column of the 96-well plate (8 wells per column, 12 columns) into 12 1.5μL Eppendorf tubes.

23. Add 128μL (0.8x) SPRI Beads to each 1.5μL Eppendorf tube. Pipette to mix and incubate for 5 minutes at room temperature.

24. Place 1.5μL tubes on DynaMag™-2 Magnet, let stand until solution in tubes is clear of beads (~5 minutes). Remove supernatant and wash beads 2x with 200uL fresh 80% EtOH. Remove all EtOH, remove tubes from magnet, and let beads dry at room temperature. DO NOT overdry beads.

25. Add 40μL EB and resuspend beads by pipet. Incubate for 5 minutes at room temperature. Place tubes back on magnet and let stand until solution is clear of beads (~5 minutes). Remove 40uL supernatant to 12 clean 1.5μL Eppendorf tubes.

26. Measure concentration of each 1.5μL Eppendorf tube with Qubit dsDNA BR Assay Kit. Normalize library concentrations and pool for sequencing.